APEX AD TECH | TECHNICAL BRIEF

Android SDK-led programmatic stack

A modular Android monetization SDK, Go SSP/auction engine and agentic control plane built as one end-to-end system.

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END-TO-END SYSTEM

ANDROID SDK - IMPLEMENTED

- Banner/MRECT, Interstitial, IAB Native 1.2, VAST 4.0 rewarded video, App Open, In-App Bidding and MRAID 3.0 over OpenRTB 2.6.
- API 21+ modular AARs; zero third-party runtime dependencies in sdk-core; lifecycle-owned ad objects, consumer rules and 40 unit/instrumentation test source files.
- Consent-aware GAID/LAT, TCF keys, GPP, US Privacy, COPPA/LMT, consent-gated first-party cohorts, real WebView UA and schain propagation.
- Optional Play Integrity leases plus Android Keystore P-256 signing; replay-resistant tracking with separate win, bill and loss lifecycle and a persistent retry queue.
- Native Android demo, AdMob adapter, bidding-token flow and a working React Native bridge for Android.

DIFFERENTIATED FORMAT WORK

- Conversational Ads for AI assistants: a newly introduced, disclosed programmatic Native format for AI-assistant chats, which most Ad Tech SDKs do not yet expose. Registered taxonomy only; no raw chat leaves the app.
- Wallet Pass Ads: Save to Google Wallet as an action layer on standard MRECT, interstitial or Native demand, with ordinary display fallback.
- Intent-to-Action Native: disclosed executable actions negotiated against capabilities actually installed in the app.

SERVER AND CONTROL PLANE

- Go SSP: first-price parallel demand fan-out, publisher/app-ads.txt registry, per-placement floors, quality gates, dynamic floor experiments, signed lifecycle callbacks and auditable auction receipts.
- Apex Intelligence: privacy-minimized contracts, real-data gates, FTRL pCTR training, model registry, human promotion and shadow-first scoring that fails closed without evidence.

AI AGENTS FOR AD TECH BOTTLENECKS

- Creative Generation Agent: converts landing pages or product briefs into multi-format drafts, reducing creative-production and setup delays.
- Creative Review Agent: versioned policy, brand/domain, landing-page and asset checks with audit evidence and manual escalation reduce approval delays.
- Campaign Copilot: evidence-grounded delivery, budget, bidding, creative and tracking diagnosis reduces buyer and AdOps troubleshooting time.
- Publisher Copilot: placement and revenue guidance plus SDK, Gradle/runtime, crash and ANR triage reduces integration and support delays, with engineering escalation.

CONTROL-PLANE SAFETY: agents are approval-gated and auditable; auctions and rendering remain deterministic.

READINESS BOUNDARY

WORKING NOW

SDK-to-auction-to-demand flow; standard rendering; privacy/trust contracts; publisher controls; creative generation/review; audit foundations.

NEXT PRODUCTION GATES

IAB OM SDK certification; fleet telemetry and kill switch; Maven/API guarantees; device, leak, ANR, fuzz, load, privacy and security validation. OMID remains honesty-gated until certified.

KEY DIFFERENTIATORS FROM COMMONLY USED AD TECH PRODUCTS

1. **SDK-direct supply**
   A first-party mobile supply path with clearer ownership, signals and auction transparency.
2. **Conversational Ads**
   A dedicated, disclosed programmatic Native format for AI-assistant chats, absent from most Ad Tech SDKs.
3. **AI workflow agents**
   Generation, review and buyer/publisher Copilots remove production, approval, operations and support bottlenecks.

GitHub reference for technical deep dives https://github.com/Madroid2/apex-ad-sdk-android